PROJECT DOCUMENTATION

Zomato Data Analysis Project

Restaurant Sales & Customer Behavior Analysis using Python

Subhadip Koley

GitHub: github.com/koleysubhadip42-netizen/Zomato-sales-and-customers-behaviour-analysis

LinkedIn: lnkd.in/p/dcha-BzS

1. Project Overview

Project name: Zomato Restaurant Sales & Customer Behavior Analysis

Developed by: Subhadip Koley

Type: Exploratory Data Analysis (EDA) in Python

Zomato is one of India's largest online food-ordering and restaurant-discovery platforms. Every restaurant listed on it generates data: ratings, votes, average cost, ordering mode and restaurant type. This project analyses that data to understand what customers order from, how they rate restaurants, how much they spend, and how ordering mode (online vs offline) affects satisfaction.

Objectives

- Clean and prepare the Zomato restaurant dataset for analysis.
- Identify the restaurant type that attracts the most customers and votes.
- Find the most common rating range across restaurants.
- Estimate the average spending for two people.
- Compare ratings between online and offline ordering modes.
- Find which restaurant types receive more offline orders so Zomato can design retention offers.

2. Business Problem

Zomato's business team wants to understand customer preferences so it can improve restaurant partnerships, pricing insights and marketing offers. The analysis is framed around six business questions:

#	Business Question	Technique Used
1	What type of restaurant do the majority of customers order from?	Count plot
2	How many votes has each type of restaurant received from customers?	GroupBy + line plot
3	What are the ratings that the majority of restaurants have received?	Histogram / value_counts(bins)
4	Zomato has observed that most couples order most of their food online. What is the average spending on each order?	Mean + histogram
5	Which mode (online or offline) has received the maximum rating?	GroupBy mean + box plot
6	Which type of restaurant received more offline orders, so that Zomato can retain customers with some good offers?	Pivot table + heatmap

Problem statements are based on the iScale 'Data Analysis Project Using Python' (Zomato) tutorial.

3. Dataset Description

The dataset is a CSV file (`Zomato data .csv`) with one row per restaurant listing. The columns used in this analysis are:

Column	Description	Type
online_order	Whether the restaurant accepts online orders (Yes / No)	Categorical
rate	Customer rating, originally stored as text like '4.1/5'	Text -> Float
votes	Number of votes given by customers	Integer
approx_cost(for two people)	Approximate cost of a meal for two	Numeric
listed_in(type)	Restaurant category (Dining, Cafes, Buffet, Other ...)	Categorical

Data quality check: the notebook confirms there are no null values and no duplicate rows, so no records had to be dropped.

4. Tools & Technologies

Tool	Purpose
Python 3	Core programming language
Pandas	Data loading, cleaning, GroupBy, pivot tables
NumPy	Numerical operations
Matplotlib	Base plotting (line plot, histogram)
Seaborn	Statistical plots (count plot, box plot, heatmap)
Jupyter Notebook / VS Code	Interactive development environment
Git & GitHub	Version control and project sharing

5. Project Workflow

The project follows a standard data-analysis pipeline, from raw data to business recommendations:

Stage	What happens
1. Data Collection	Import libraries and load the CSV into a DataFrame with pd.read_csv().
2. Data Cleaning	Convert the 'rate' column from text ('4.1/5') to a float using a custom function.
3. Data Understanding	Use info(), isnull().sum() and duplicated().sum() to verify data quality.
4. EDA & Visualization	Answer each business question using the most suitable chart (count, line, histogram, box, heatmap).
5. Insight Extraction	Read each chart / table and note the finding as a comment in the notebook.
6. Recommendations	Translate findings into actions for Zomato (offers and retention ideas).

6. Step-by-Step Implementation

Step 1 - Import libraries and load data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Zomato data .csv")
print(df.head())
```

Step 2 - Clean the 'rate' column

Ratings are stored like 4.1/5. The function splits on '/' and keeps the numerator as a float.

```
def rate(value):
    value = str(value).split("/")
    value = value[0]
    return float(value)

df["rate"] = df["rate"].apply(rate)
df["rate"] = df["rate"].astype(float)
```

Step 3 - Data understanding

```
print(df.info())
print("Total null values:", df.isnull().sum().sum())
print("Total duplicate values:", df.duplicated().sum())
```

Result: no missing values and no duplicates.

Q1 - Which restaurant type gets the most orders?

```
sns.countplot(x=df["listed_in(type)"])
plt.xlabel("types of restaurant")
plt.ylabel("Count of Orders")
plt.show()
```

Finding: The majority of restaurants fall under the Dining type.

Q2 - Votes per restaurant type

```
every_restaurant_vote = df.groupby("listed_in(type)")["votes"].sum()
print(every_restaurant_vote)
plt.plot(every_restaurant_vote, c="Red", marker="o")
plt.xlabel("Types Of Restaurants")
plt.ylabel("Votes")
```

Finding: Dining receives the most votes, followed by Other, then Cafes, then Buffet.

Q3 - Most common rating range

```
print(df["rate"].value_counts(bins=5))
plt.hist(df["rate"], bins=5)
plt.title("Rating distribution")
plt.show()
```

Finding: Most restaurants are rated between 3.4 and 3.8.

Q4 - Average spending for two

```
avg = df["approx_cost(for two people)"].mean()
print("Average Couple Spending:", avg)
sns.histplot(df["approx_cost(for two people)"], bins=20)
plt.show()
```

Finding: The average couple spends a little over 400 (rupees) per order. The exact value is printed by the notebook.

Q5 - Online vs offline ratings

```
print(df.groupby("online_order")["rate"].mean())
sns.boxplot(x=df["online_order"], y=df["rate"], data=df)
plt.show()
```

Finding: Restaurants that accept online orders have a higher average rating than those that do not.

Q6 - Offline orders by restaurant type

```
pivot_table = df.pivot_table(index="listed_in(type)",
                             columns="online_order", aggfunc="size")
print(pivot_table)
sns.heatmap(pivot_table)
plt.title("customer behaviour")
plt.show()
```

Finding: Dining restaurants have the most offline (No) orders, while Cafes lean more towards online orders. Dining customers are therefore the best target for retention offers.

7. Key Insights

The six business questions produced the following findings. Each is stated together with how it should be read, so that the numbers are not over-interpreted.

Question	Insight	How to read it
Most popular type	Dining restaurants make up the largest share of listings.	Dining is the biggest segment of the platform.
Votes	Dining > Other > Cafes > Buffet.	Total votes grow with the number of restaurants, so part of Dining's lead is volume.
Ratings	Most restaurants fall in the 3.4 - 3.8 band.	Ratings are tightly clustered, so small gains separate restaurants.
Spending	Average cost for two is just above 400.	This is a mean and can be pulled up by a few expensive restaurants.
Ordering mode	Online-order restaurants have higher average ratings.	This is an association, not proof that online ordering raises ratings.
Offline orders	Dining restaurants have the highest offline-order count; Cafes lean online.	Dining is the natural segment for retention offers.

8. Challenges & Fixes

Challenge	Cause	Solution
AttributeError: module 'pandas' has no attribute 'Series'	A personal file named pandas.py in the Downloads folder shadowed the real pandas library.	Rename or delete pandas.py (and its __pycache__ folder), then restart the kernel.
Rating stored as text ('4.1/5')	The source data keeps the rating and its scale together.	Custom rate() function followed by astype(float).
Filename contains a space	The file is named 'Zomato data .csv'.	Use the exact name in code, or rename it to Zomato_data.csv and update read_csv().

Lesson learned: never name a personal script after a library (pandas.py, numpy.py, random.py), and keep dataset filenames free of spaces.

9. Business Recommendations

- **Target Dining customers with loyalty offers.** They place the most offline orders and give the most votes, so retention offers here reach the largest group.
- **Encourage restaurants to enable online ordering.** Online-enabled restaurants show higher average ratings; Zomato can offer onboarding support to offline-only partners.
- **Use price-aware promotions.** With average spend for two around 400, discounts and combos near this range are likely to convert well.

- **Help low-rated restaurants improve.** Since most ratings sit in the 3.4 - 3.8 band, small quality improvements can move a restaurant above the crowd.
- **Grow the Cafes segment online.** Cafes already lean towards online orders, so digital promotions suit them best.

10. Limitations & Future Scope

Limitations

- The analysis is descriptive: it shows patterns, not causes.
- The dataset is a snapshot, so trends over time cannot be studied.
- A higher online-order rating does not prove that online ordering causes better ratings.
- Total votes depend on how many restaurants each type has, so votes per restaurant would be a fairer comparison.
- Only five columns were used; location and cuisine were not analysed.

Future scope

- Build an interactive Power BI / Excel dashboard of these findings.
- Add location and cuisine analysis for deeper segmentation.
- Apply statistical tests (t-test) to confirm online vs offline rating differences.
- Compare average votes per restaurant instead of total votes.
- Train a machine-learning model to predict restaurant ratings.

11. Conclusion

This project demonstrates a complete data-analysis workflow in Python, from loading and cleaning raw data to visual exploration and business recommendations. The analysis shows that Dining restaurants lead in votes and offline orders, ratings cluster between 3.4 and 3.8, and online-enabled restaurants enjoy better ratings.

These insights can help Zomato design targeted offers and improve customer retention. Because the analysis is descriptive, the next step is to confirm the findings with statistical tests and richer data such as location and cuisine.

12. About the Author

Subhadip Koley